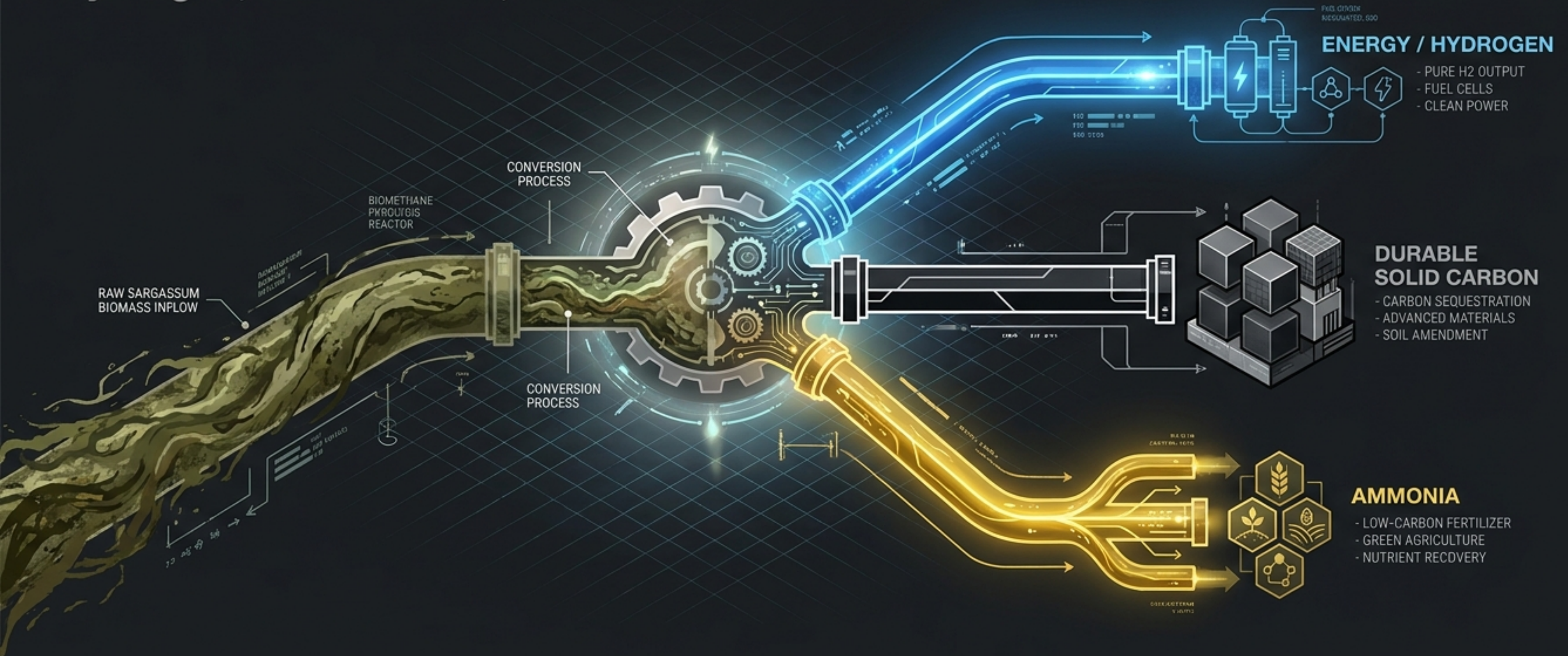
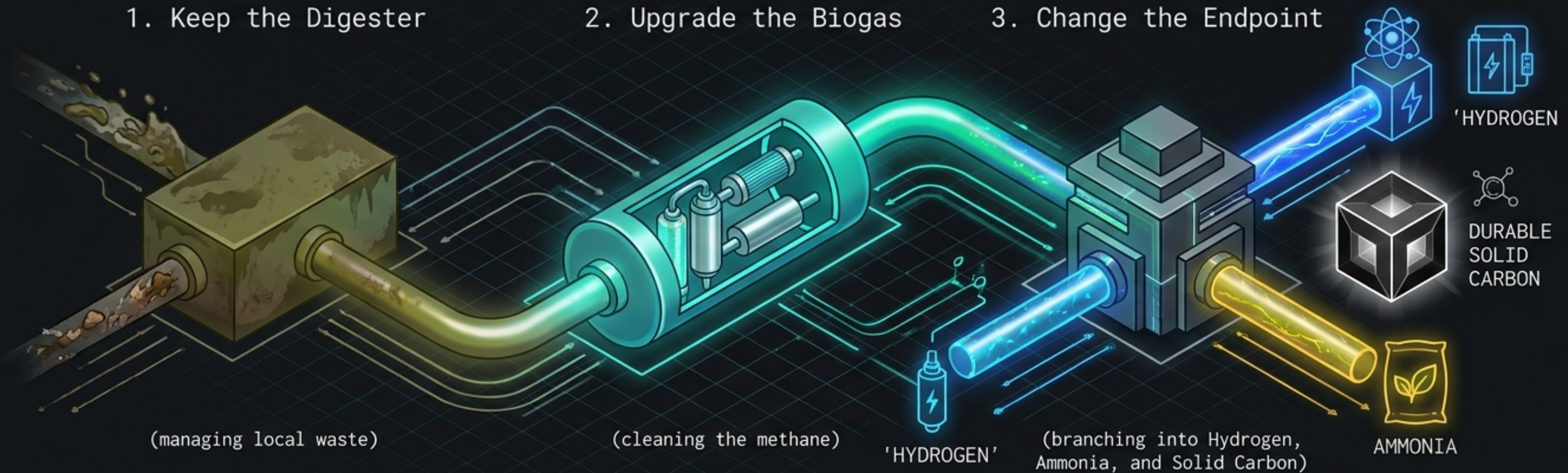


From Nuisance to Low-Carbon Fertilizer

A Biomethane Pyrolysis Pathway for Coastal Resilience, Hydrogen, Solid Carbon, and Ammonia.

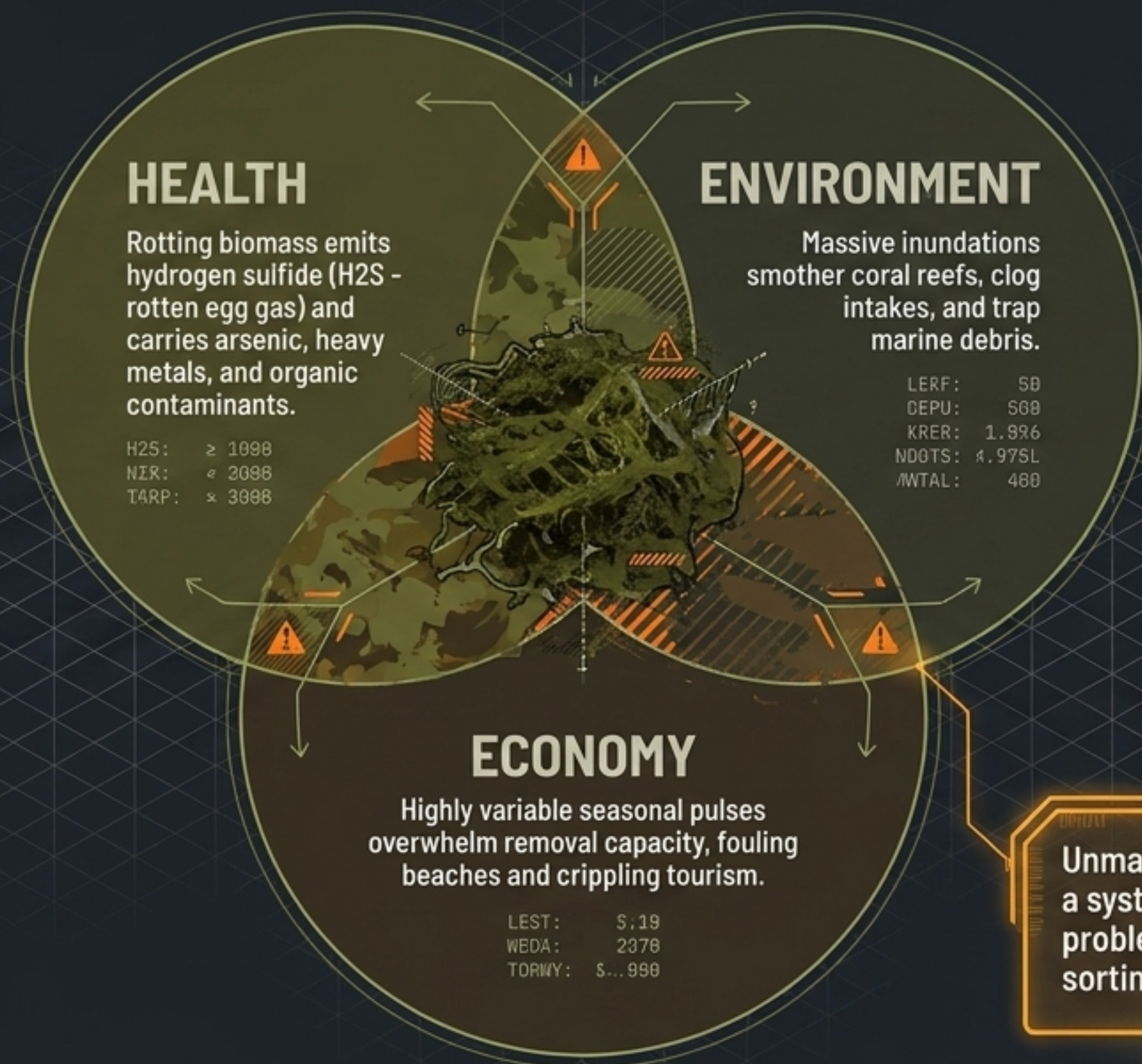


A Bolt-On Upgrade for Existing Coastal Biogas



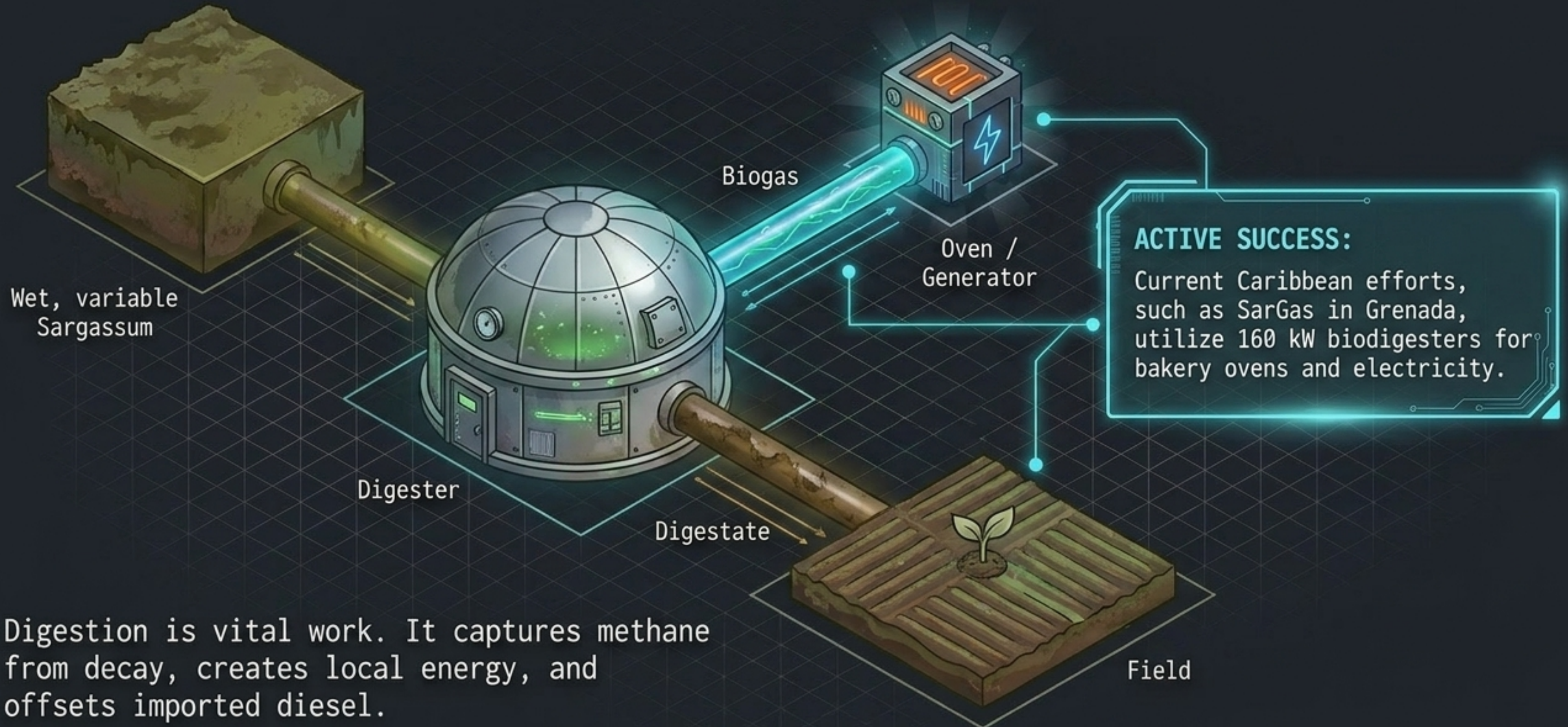
Most Sargassum projects stop at heat, electricity, or raw biofertilizer. By adding a biomethane pyrolysis branch, we transform the methane fraction into conventional ammonia fertilizer and solid carbon—leaving the combustion paradigm behind.

Sargassum is a Practical Crisis Before it is a Climate Problem

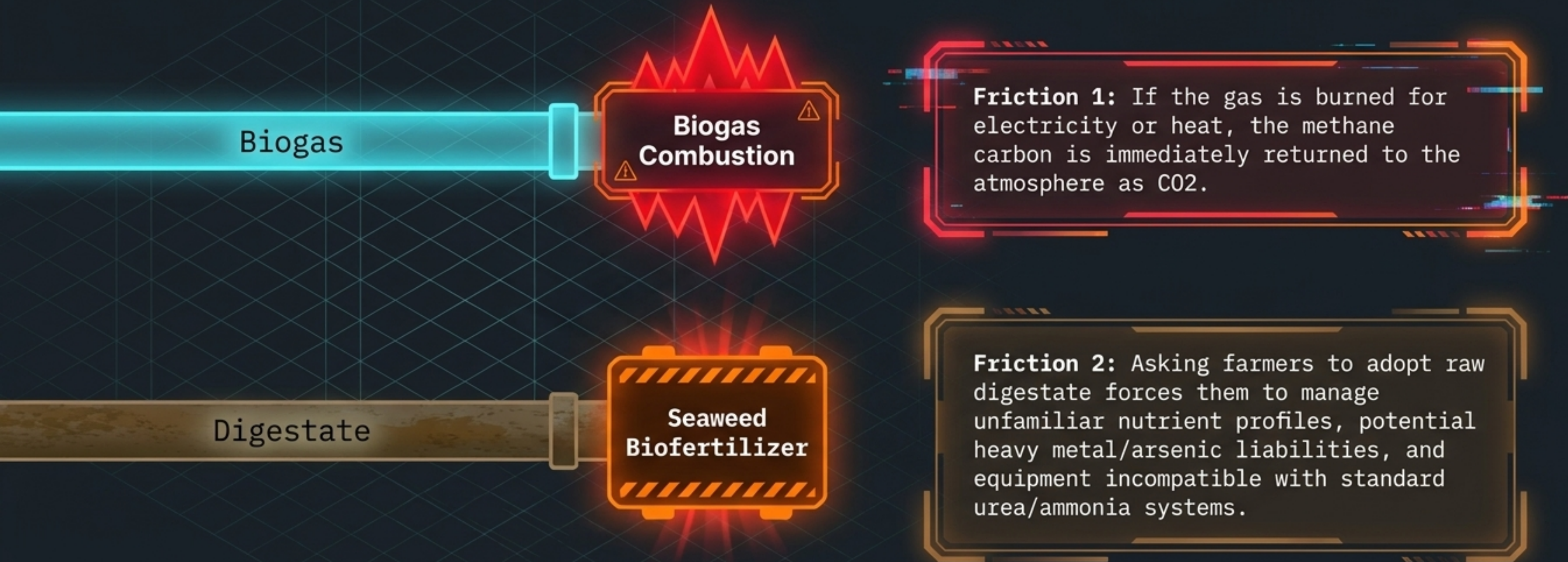


Unmanaged beach accumulation requires a system that treats biomass as a problematic wet feedstock, requiring sorting and contaminant management.

Anaerobic Digestion is the Right Foundation



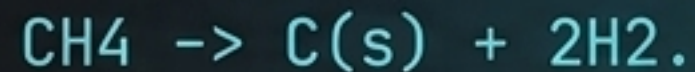
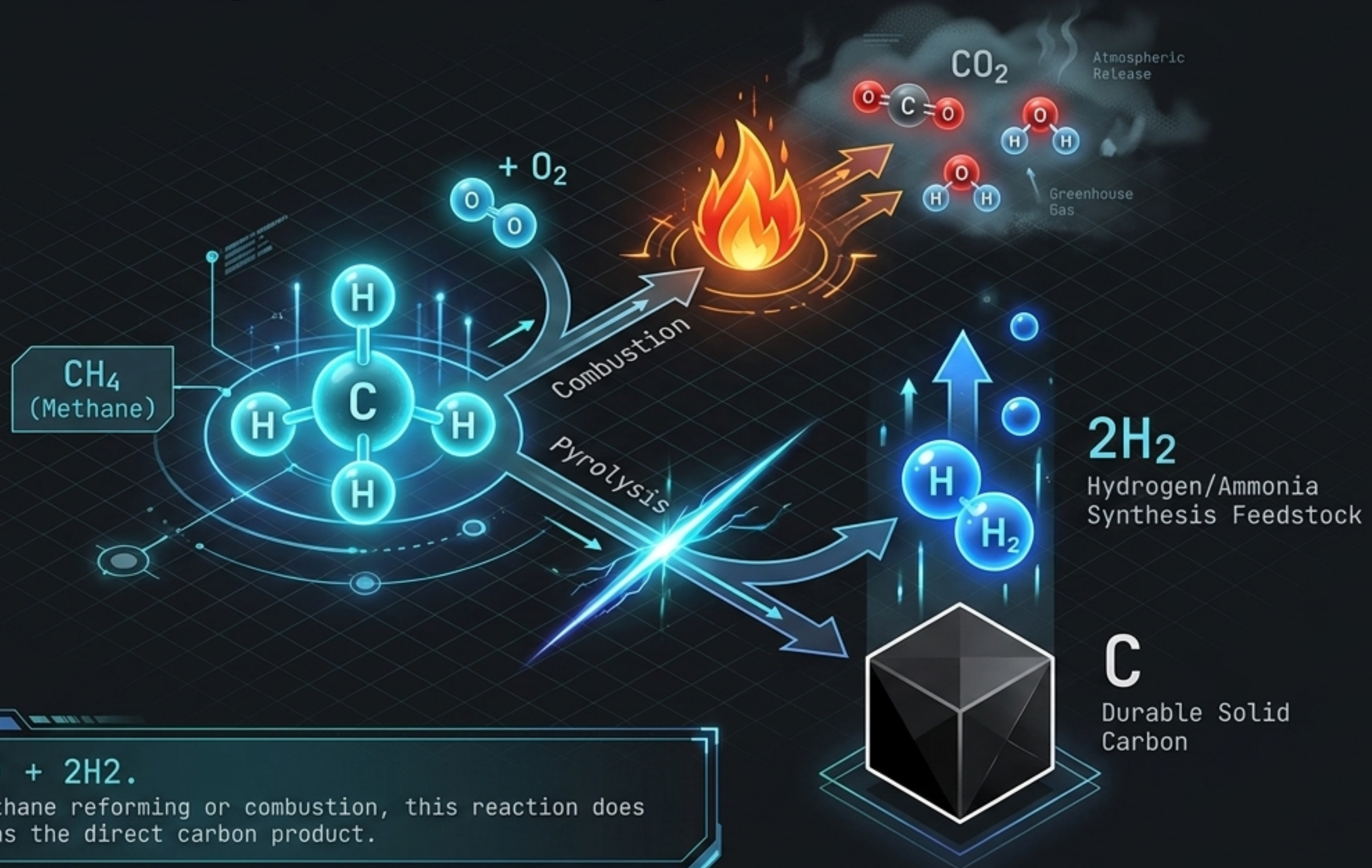
Combustion and Raw Digestates Leave Value on the Table



The 7-Step Biomethane Pyrolysis Pathway

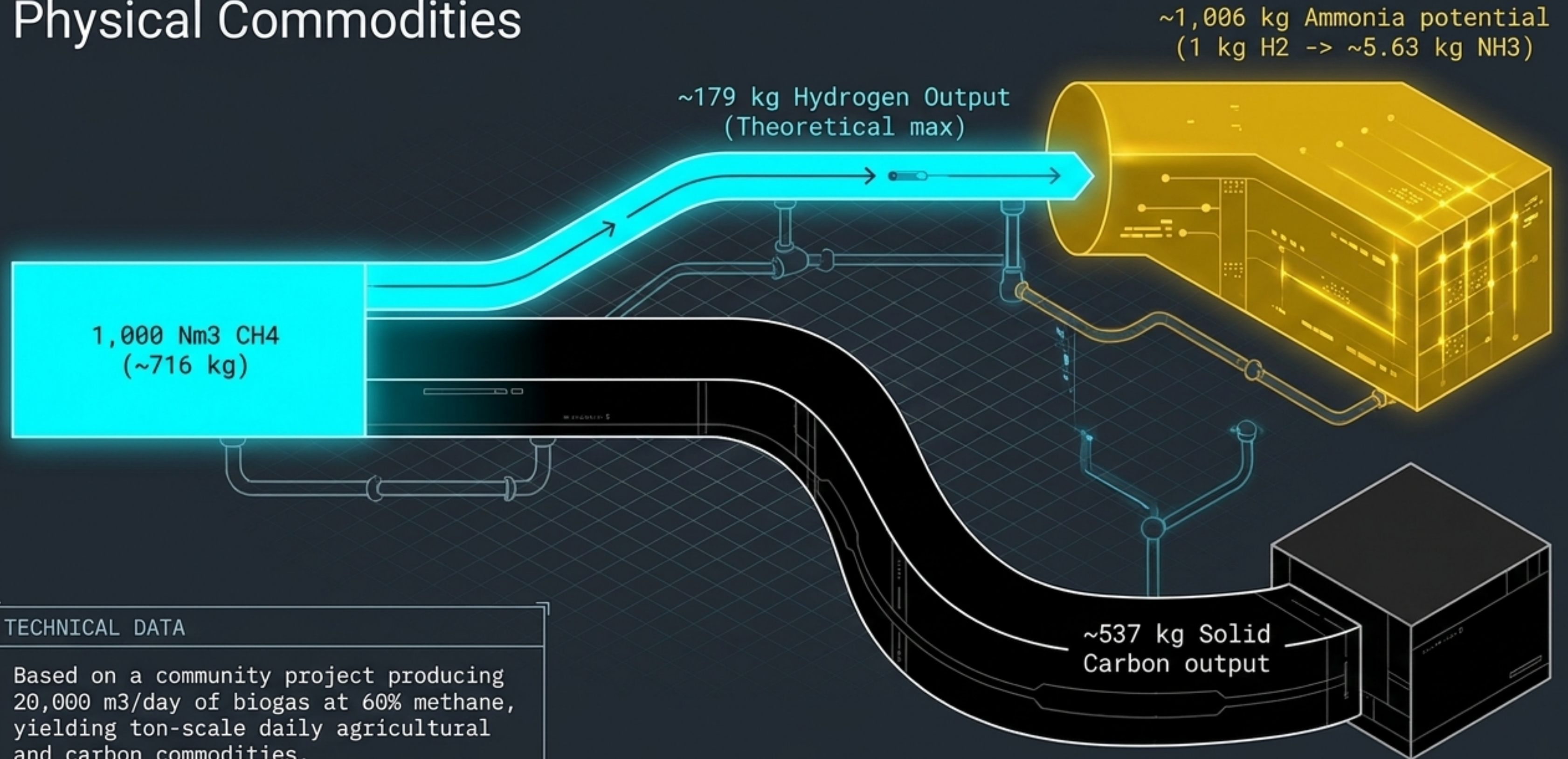


Cleaving the Molecule: Pyrolysis vs. Combustion



Unlike steam methane reforming or combustion, this reaction does not create CO_2 as the direct carbon product.

Theoretical Yield: Converting Gas to Physical Commodities



TECHNICAL DATA

Based on a community project producing 20,000 m³/day of biogas at 60% methane, yielding ton-scale daily agricultural and carbon commodities.

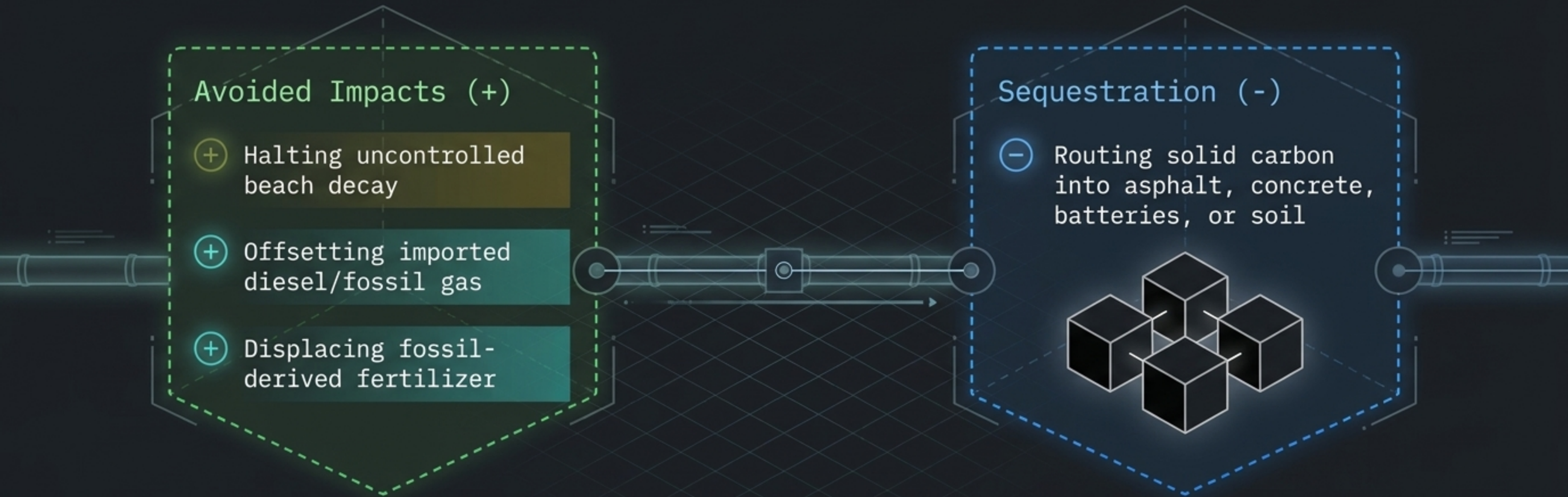
Stop Asking Farmers to Change; Change the Chemistry Instead



The Paradigm Shift: Combustion vs. Pyrolysis

	CURRENT BIOGAS PATHWAY	PROPOSED PYROLYSIS PATHWAY
CARBON FATE	Emitted to atmosphere as CO ₂	Captured as physical solid carbon
MAIN ENERGY	Combusted for heat/electricity	Upgraded to clean Hydrogen 
MAIN FERTILIZER	Raw, variable seaweed digestate	Standardized Haber-Bosch Ammonia 
FARMER ADOPTION	High friction (new logistics, liability)	Zero friction (existing infrastructure)
CLIMATE NARRATIVE	Avoided emissions (replacing diesel)	Durable carbon / net-negative potential 

Systemic Environmental Logic and Carbon Boundaries



CAUTION: This pathway is not automatically carbon-negative. True net-negative accounting requires strict management of the biogenic CO₂ fraction removed during upgrading, and ensuring the final solid carbon enters a genuinely durable market.

Engineering Reality: Anticipating Operational Friction

Industrial Risk



Salt, Sulfur & H₂S
(Highly corrosive)



Feedstock Variability
(Seasonal pulses)



Arsenic/Heavy Metals
(Agronomic contamination)



Economics of Small Scale
(Haber-Bosch is massive)

Engineered Mitigation

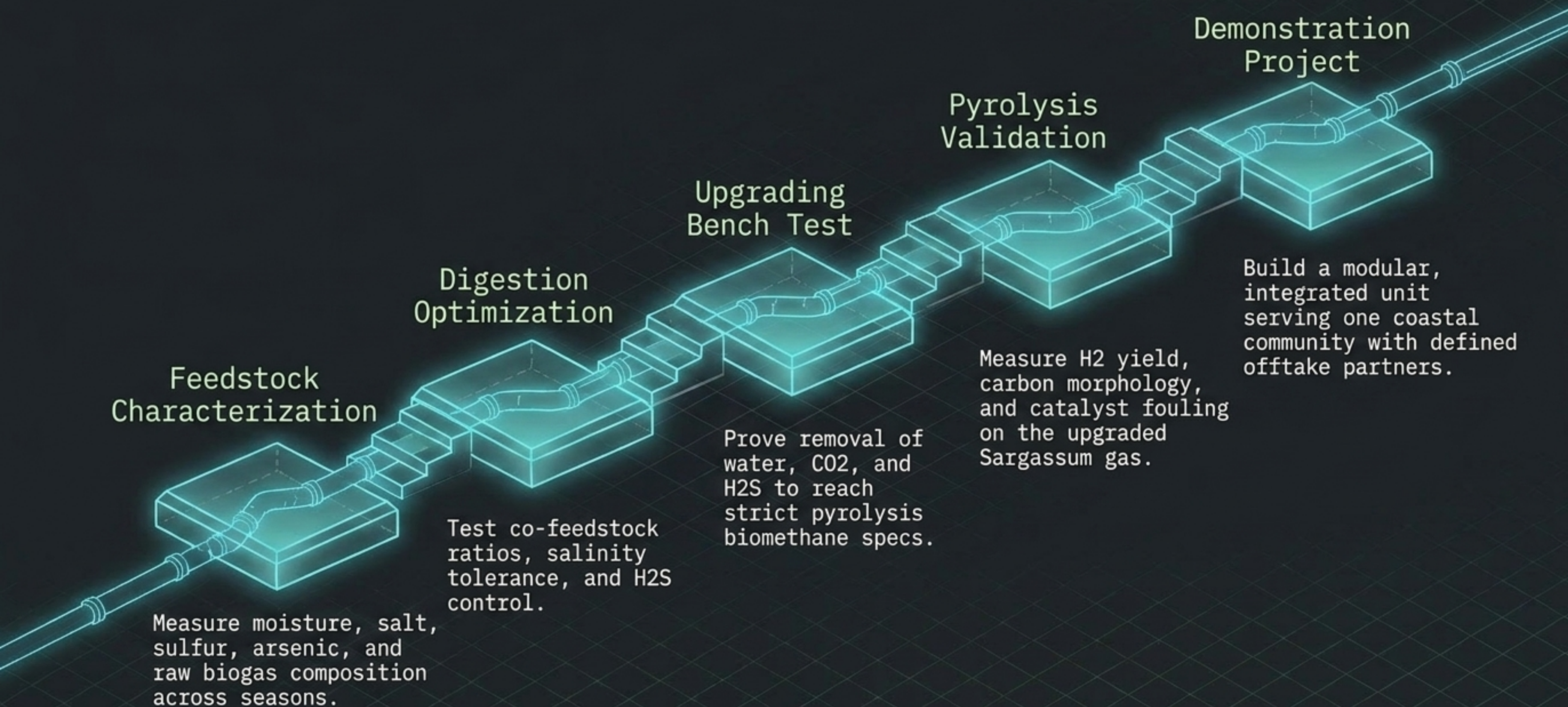
Strict gas scrubbing, H₂S monitoring, and upgrading to pure biomethane to protect pyrolysis catalysts.

Co-digestion (food waste, manure) to stabilize biology and flexible modular throughput.

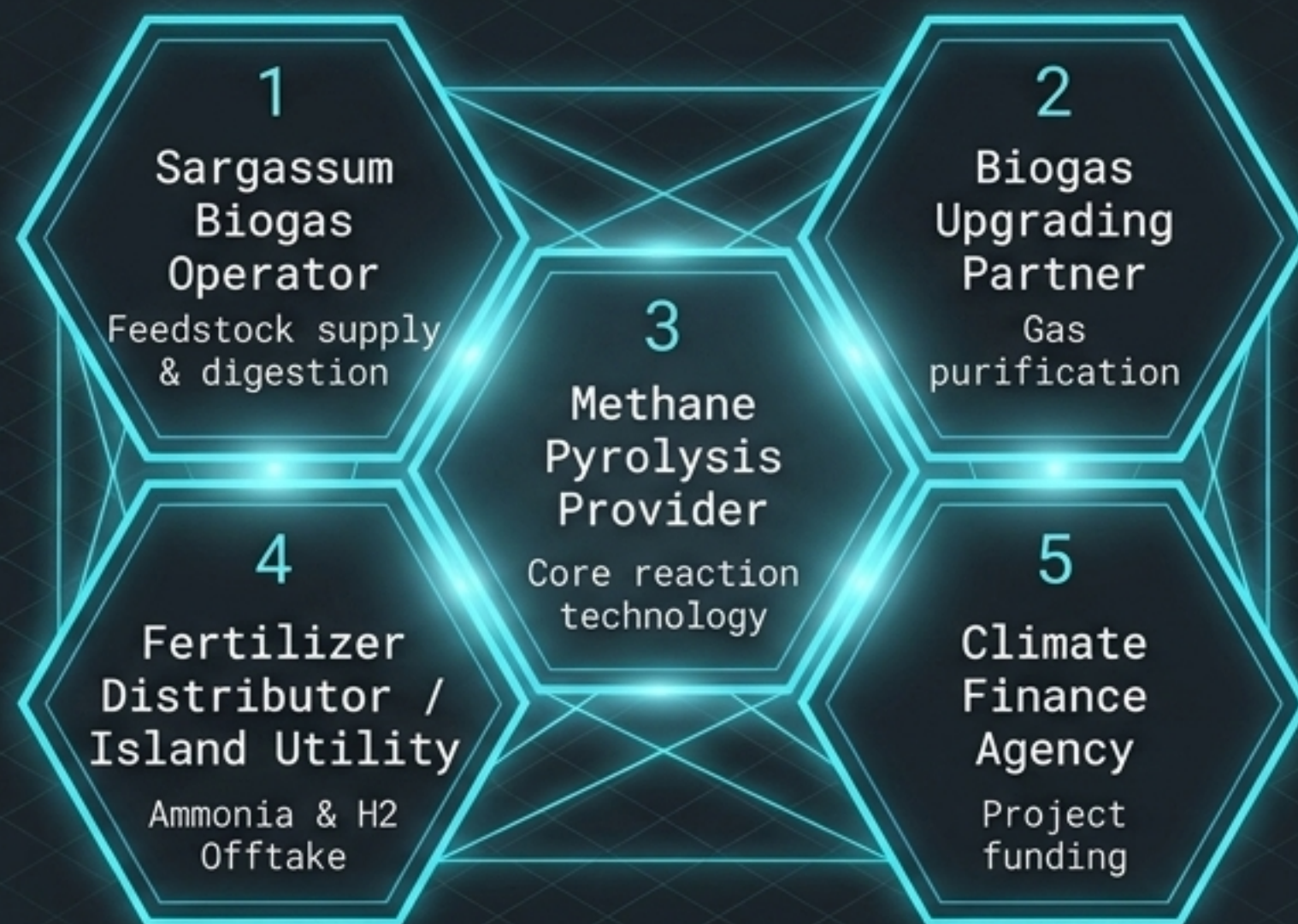
Routine testing; route carbon to non-food durable materials; keep digestate secondary.

Begin with H₂ local use/fuel cells, utilize modular regional hub-and-spoke ammonia blending.

A Staged Path from Bench Test to Demonstration



Assembling the Integration Ecosystem



The immediate next step: Initiate a joint feasibility study to run a gas-spec and product-spec bench test on upgraded Sargassum biomethane. Connecting coastal cleanup to global food security.